Overview. SmartNICs accelerate datacenter systems by offloading networking, storage, and application functions from host CPUs, improving performance and cost efficiency. In storage systems, SmartNIC-based Just a Bunch of Flash (JBOF) architectures realize these benefits by using onboard processors as storage controllers to expose flash storage over the network, eliminating the need for a separate host CPU and host memory. During my Ph.D., I aim to explore how to rearchitect storage clusters in disaggregated systems using SmartNIC-based JBOFs.

Research Experiences. During my M.S. at UNIST, I worked with **Prof. Youngbin Im** on FlexServer[1], a research project focused on optimizing content servers that deliver files from storage to clients over the network. Reading files from storage can stall the server because the process must wait until the file data has been loaded into the TCP send buffer before continuing with other work. To address this, I implemented a transmission system built on multi-tier caching that prefetches hot files from NVMe SSDs into host DRAM and SmartNIC memory. I implemented this system on both BlueField and Agilio SmartNICs. A key challenge was preserving correct TCP behavior while adhering to a zero-copy design. I reorganized the TCP send buffer to handle data according to its location: requesting the SmartNIC to transmit data stored in its memory, passing physical addresses to the NIC for data in host memory, and using an optimized I/O engine to read data from NVMe SSDs into the TCP send buffer. FlexServer achieved up to 351 Gbps, outperforming the state-of-the-art baseline, IO-TCP[2], by 4.4×. I am the first author of the resulting paper, which is currently under submission. However, the pipeline still used a bounce buffer to stage data from NVMe devices before network transmission. This led me to explore FPGA-based SmartNICs as a way to eliminate this staging step through direct data transfers between NVMe devices and the NIC.

At MangoBoost, I worked on software stacks for NVMe/TCP accelerators built on FPGA-based SmartNICs. As a coauthor of BoostX-NTI[3], accepted at ISCA 2026, I developed its PCIe interface and NVMe-compliant control plane. Since managing complex state in hardware is challenging, we ran the control plane on an onboard Arm processor to handle NVMe register configuration and admin commands, while keeping performance-critical NVMe I/O processing in FPGA logic. A key challenge was coordinating these components to preserve correct NVMe behavior during network faults, for which I developed fault-handling mechanisms. The accelerator exposed remote NVMe/TCP storage as a local NVMe/PCIe device, supporting Windows and Linux hosts and earning UNH-IOL certification and a listing on the UNH-IOL NVMe Integrators List[4]. In this project, hardware offloading focused on NVMe read and write commands with a simple data path: transferring data via DMA using host memory addresses. Had we used BlueField, data would have needed to be staged in Arm memory before transmission by the NIC.

Motivation. My experience with FPGA-based SmartNICs showed me how their flexibility enables low-level optimizations. For example, P2PDMA can move data directly between NVMe SSDs and a NIC, bypassing host DRAM. Yet despite vendors' description of FPGAs as "software-like hardware," an FPGA's *development velocity* is substantially lower than that of software development. This tradeoff led me to turn my attention toward SoC-based SmartNICs. While my company's work remained centered on FPGA-based systems, my research interests increasingly shifted toward SoC-based designs. As BlueField has evolved, its expanding hardware capabilities have made it increasingly compelling for storage. I believe that mapping storage functions to a SmartNIC-based JBOF according to its resource characteristics could yield a practical system while preserving the features of existing storage systems. I want to pursue a Ph.D. to systematically investigate how storage systems can be designed around the capabilities and constraints of SmartNICs.

Rethinking Storage Software for SmartNIC-Based JBOFs. LLM training and inference workloads often involve large objects, such as model checkpoints and weights, creating a need for storage systems that can serve these objects at high throughput. I am interested in extending SmartNIC-based storage beyond the small-object key-value workloads evaluated in LEED[5] to support these demands. A key challenge in

serving large objects is limited memory bandwidth, compounded by the constraints imposed by SmartNICs' wimpy cores. BlueField-3 is expected to offer roughly four times the memory bandwidth of BlueField-2, and BlueField-4 roughly three times that of BlueField-3[6;7], suggesting greater potential for memory-intensive workloads. Nevertheless, BlueField-3's measured memory throughput suggests that staging data in onboard memory between NVMe I/O and network transfers remains challenging[8]. A key challenge is therefore how to allocate memory bandwidth among storage functions. For example, under a 9:1 read/write request mix, a scheduler may allocate more bandwidth to higher-priority operations rather than divide it evenly.

Meanwhile, BlueField-3 combines onboard Arm cores and DPA cores with distinct characteristics. For example, DPA cores can access a dedicated region of onboard memory and efficiently handle data transfers for parallel workloads whose working sets fit within their caches. Their support for many hardware threads per core also provides substantial parallelism[8]. I envision a storage system that assigns functions to these heterogeneous processors according to their memory-access, cache, and parallelism characteristics.

Why UW–Madison. I am particularly interested in working with **Prof. Ming Liu** because of his Programmable Disaggregated Storage project. While working on BoostX-NTI, I read his research on NVMe-oF and SmartNIC-based storage with great interest, including ntprof[9], SANarch[10], Gimbal[11], and LEED[5]. I would also like to work with **Prof. Andrea Arpaci-Dusseau**. I believe the challenges addressed in her work on uFS[12], such as file ownership management and dynamic load balancing, could also arise in disaggregated storage systems built on SmartNICs. Finally, I would welcome the opportunity to work with **Prof. Matt Sinclair**. I am interested in whether and how chiplet-based CPU features, including Sub-NUMA Clustering (SNC), affect data movement involving not only GPUs but also NICs and NVMe SSDs, and how software stacks should evolve to account for these effects.

My experience with SmartNICs, DPDK/SPDK, RDMA/TCP, and disaggregated storage has prepared me to contribute to systems research at UW–Madison. My long-term goal is to lead research that turns new storage architectures into deployed datacenter systems, whether in academia or at an industrial research lab. Both paths require defining problems from first principles rather than following product roadmaps, and I hope to develop that ability through a Ph.D.

References

[1] Dongu Kim, Seyeong Park, Seyeon Kim, Sangtae Ha, and Youngbin Im. FlexServer: A High-Performance Content Server Leveraging Asynchronous I/O and SmartNICs. Preprint, 2024. URL `https://ddubbie.github.io/assets/pdf/preprint-flexserver.pdf`.

[2] Taehyun Kim, Deondre Martin Ng, Junzhi Gong, Youngjin Kwon, Minlan Yu, and KyoungSoo Park. Rearchitecting the TCP Stack for I/O-Offloaded Content Delivery. In *Proceedings of the 20th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2023.

[3] Hamin Jang, Jinha Jeong, Wonseok Lee, Jun Heo, Jongcheon Lee, Hyunjae Chu, Taehyun Kim, Youngwoo Jeong, Dongu Kim, Heetaek Jeong, Changsu Kim, Dongup Kwon, and Jangwoo Kim. BoostX-NTI: Fast, Scalable and Flexible Storage Architecture with NVMe/TCP Initiator Acceleration. In *Proceedings of the 53rd ACM/IEEE International Symposium on Computer Architecture (ISCA)*, 2026. doi: 10.1109/ISCA66397.2026.00081.

[4] University of New Hampshire InterOperability Laboratory. NVMe Integrators List, n.d. URL `https://www.iol.unh.edu/registry/nvme`.

[5] Zerui Guo, Hua Zhang, Chenxingyu Zhao, Yuebin Bai, Michael Swift, and Ming Liu. LEED: A Low-Power, Fast Persistent Key-Value Store on SmartNIC JBOFs. In *Proceedings of the ACM SIGCOMM 2023 Conference*, 2023.

[6] NVIDIA. Power the Next Wave of Applications with NVIDIA BlueField-3 DPUs. NVIDIA Technical Blog, 2023. URL `https://developer.nvidia.com/blog/power-the-next-wave-of-applications-with-nvidia-bluefield-3-dpus/`.

[7] Kyle Aubrey. Inside the NVIDIA Vera Rubin Platform: Six New Chips, One AI Supercomputer. NVIDIA Technical Blog, January 2026. URL `https://developer.nvidia.com/blog/inside-the-nvidia-rubin-platform-six-new-chips-one-ai-supercomputer/`.

[8] Xuzheng Chen, Jie Zhang, Ting Fu, Yifan Shen, Shu Ma, Kun Qian, Lingjun Zhu, Chao Shi, Yin Zhang, Ming Liu, and Zeke Wang. Demystifying Datapath Accelerator Enhanced Off-Path SmartNIC. In *Proceedings of the 32nd IEEE International Conference on Network Protocols (ICNP)*, 2024.

[9] Yuyuan Kang and Ming Liu. Understanding and Profiling NVMe-over-TCP Using ntprof. In *Proceedings of the 22nd USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2025.

[10] Chendong Wang, Joontaek Oh, and Ming Liu. Co-Designing Traffic Control with NVMe-oF for Disaggregated Storage: A Comparative Study of Switched and Switchless SAN Architectures. In *Proceedings of the 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2026.

[11] Jaehong Min, Ming Liu, Tapan Chugh, Chenxingyu Zhao, Andrew Wei, In Hwan Doh, and Arvind Krishnamurthy. Gimbal: Enabling Multi-tenant Storage Disaggregation on SmartNIC JBOFs. In *Proceedings of the ACM SIGCOMM 2021 Conference (SIGCOMM '21)*, pages 106–122, 2021. doi: 10.1145/3452296.3472940.

[12] Jing Liu, Anthony Rebello, Yifan Dai, Chenhao Ye, Sudarsun Kannan, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. Scale and Performance in a Filesystem Semi-Microkernel. In *Proceedings of the 28th ACM Symposium on Operating Systems Principles (SOSP)*, 2021.